



Multiplicative relationships in context

Task A: Multiplicative relationships in context

1) Fill in the table knowing the following multiplicative relationships:

a) Number of eggs $\times 75 =$ Sugar

b) Flour is twice as many grams as butter

c) Butter $\times 1.5 =$ Sugar

Butter (g)	Flour (g)	Eggs	Sugar (g)
100	200	2	150
400	800	8	600
500	1000	10	750

2) Most bread doughs start with flour and water. To make chapati, for every two cups of flour, you add one cup of water. To make pizza base, for every three cups of flour, you add two cups of water.

In each bowl, is it chapati dough, pizza base dough, or neither?

Dough	Cups of flour	Cups of water
1 chapati	2	1
4.5 chaptis	9	4.5
6 chaptis	12	6
9 chaptis	18	9
12 chaptis	24	12

Dough	Cups of flour	Cups of water
1 pizza	3	2
3 pizzas	9	6
4 pizzas	12	8
6 pizzas	18	12
8 pizzas	24	16

a)  9 cups of flour
6 cups of water
Pizza

b)  12 cups of flour
8 cups of water
Pizza

c)  12 cups of flour
6 cups of water
Chapati

d)  18 cups of flour
12 cups of water
Pizza

e)  24 cups of flour
12 cups of water
Chapati

f)  24 cups of flour
18 cups of water
Neither



Task A: Multiplicative relationships in context

3) To make the perfect meringue, for every three eggs you add 150 g sugar. What is the minimum needed to add to each mixture below to make the meringue perfect?

a)



200 g sugar
6 eggs

*Recipe x2:
300 g sugar
6 eggs*

Add 100g sugar

b)



450 g sugar
7 eggs

*Recipe x3:
450 g sugar
9 eggs*

Add 2 eggs

c)



400 g sugar
9 eggs

*Recipe x3:
450 g sugar
9 eggs*

Add 50 g sugar

d)



500 g sugar
9 eggs

*Recipe x4:
600 g sugar
12 eggs*

*Add 200 g sugar
Add 3 eggs*

Another solution for d)

*Sugar $\div$ 50 = eggs
500 g sugar
10 eggs*

Add 1 egg

e)



500 g sugar
10 eggs

No change. It is perfect!

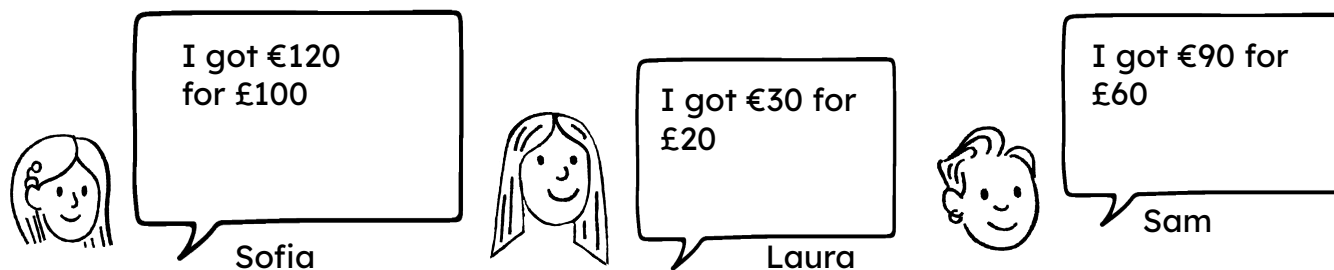


Task B: Converting money

1) Fill in the currency conversion table using the following ratio:

Euros (€)	USA dollars (\$)
1	1.08
11	11.88
20	21.60
0.5	0.54
50	54
150	162

2) A group of three friends are going on holiday. They all exchange some pounds for euros.



a) Who got the best deal?

Laura and Sam got the best deal

Sofia		Laura		Sam	
Pounds (£)	Euros (€)	Pounds (£)	Euros (€)	Pounds (£)	Euros (€)
100	120	20	30	60	90
$100 \times 1.2 = 120$		$20 \times 1.5 = 30$		$60 \times 1.5 = 90$	
$120 \div 100 = 1.2$		$30 \div 20 = 1.5$		$90 \div 60 = 1.5$	
Pounds $\times 1.2$ = Euros		Pounds $\times 1.5$ = Euros		Pounds $\times 1.5$ = Euros	

b) Andeep decided to come along too and exchanged €110 for £100. Whose deal was it most similar to?

$110 \div 100 = 1.1$ this is most similar to Sofia

